

Outline

Pseudo-Random Number Generators ✓

Generating Uniform and Non-Uniform Random Variables ✓

Inverse Transform Method ✓

Acceptance-Rejection Method ✓

Box-Muller Transformation ✓ →

Ratio-of-Uniforms Method ✓

Theory

Practical Algorithm

Example: Standard Normal via Ratio-of-Uniforms

Diagnosing Randomness

Unit Summary

The Ratio-of-Uniforms Idea

Theorem (Kinderman-Monahan, 1977)

Let $h(x) \geq 0$ be proportional to the target density $f(x)$ (i.e. $f(x) = h(x) / \int h$). Define the planar region

$$C = \left\{ (u, v) : x = \frac{v}{u}, 0 \leq u \leq \sqrt{h(x)}, b^- \leq v \leq b^+ \right\}.$$

If (U, V) is drawn **uniformly** over C , then $X = V/U$ has density f .

$$f(x) = \frac{h(x)}{\int h(t) dt}$$

norm. const

$$\int f(x) dx = 1$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{\pi}} e^{-x^2/2}$$

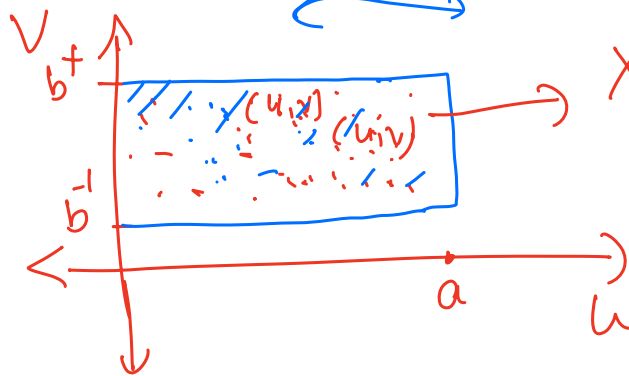
$$\Rightarrow \frac{h(x) \geq 0}{\uparrow} \rightarrow f(x) = \frac{h(x)}{\int h(t) dt} = \frac{e^{-x^2/2}}{\int_{-\infty}^{\infty} e^{-t^2/2} dt} = \frac{e^{-x^2/2}}{\sqrt{\pi}}$$

pdf

$$\Rightarrow \checkmark C = \left\{ (u, v) : x = \frac{v}{u}, \quad \begin{array}{l} 0 \leq u \leq \sqrt{n(x)} \\ \bar{b} \leq v < b^+ \end{array} \right\}$$

$$U \sim U[\underbrace{0, \sqrt{n(x)}}_{\leftarrow}], \quad V \sim U[\underbrace{\bar{b}^+, b^+}_{\leftarrow}]$$

$$\text{Then } \underline{X = \frac{V}{U}} \sim \underline{f}$$



$$X_1 = \frac{V_1}{U_1}, \quad X_2 = \frac{V_2}{U_2}$$

$$f: \quad X_n = \frac{V_n}{U_n} \sim f$$

The Ratio-of-Uniforms Idea

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If (U, V) is drawn **uniformly** over C , then $X = V/U$ has density f .

- Beautiful because it needs f only up to a normalizing constant – exactly like acceptance–rejection.

$$\int h(t) dt$$

- Sampling uniformly over the (often irregularly shaped) region C is done by enclosing C in a bounding rectangle and rejecting points that fall outside C .

Constructing the Bounding Rectangle

For a density with a bounded, unimodal $h(x)$, define

$$a = \sup_x \sqrt{h(x)}, \quad b^+ = \sup_x x \sqrt{h(x)},$$

Then $C \subset [0, a] \times [b^-, b^+]$, a simple bounding rectangle.

$$\inf_x x \sqrt{h(x)} = b^- \leq V \leq b^+ = \sup_x x \sqrt{h(x)}$$

Handwritten notes: $u \leq \sqrt{h(x)} \Rightarrow u x \leq x \sqrt{h(x)}$

$$b^- = \inf_x x \sqrt{h(x)}$$

$$0 \leq u \leq \sqrt{h(x)} \Rightarrow C \subset [0, a] \times [b^-, b^+]$$

we want

$$u \leq \sqrt{h(x)} \Rightarrow a = \sup_x \sqrt{h(x)}$$

$$\forall x \quad b^- \leq V \Rightarrow x = \frac{V}{u} \Rightarrow V = u x$$

$$b^+ = \sup_x x \sqrt{h(x)}$$

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$$U < \sqrt{0.5 h(x)}$$

Algorithm: Ratio-of-Uniforms Sampling

- 
1. Generate $U \sim \text{Uniform}(0, a)$ and $V \sim \text{Uniform}(b^-, b^+)$.
 2. Compute $X = V/U$.
 3. **Accept** X if $U \leq \sqrt{h(X)}$; otherwise **reject** and return to Step 1.

Worked Example: Standard Normal

Take $h(x) = e^{-x^2/2}$ (proportional to the standard Normal density).

$$a = \sup_x \sqrt{h(x)} = \sup_x e^{-x^2/4} = 1 \quad (\text{at } x = 0),$$

$$b^+ = \sup_x x e^{-x^2/4}. \quad \text{Setting the derivative to zero:}$$

$$\frac{d}{dx} [x e^{-x^2/4}] = e^{-x^2/4} \left(1 - \frac{x^2}{2}\right) = 0 \Rightarrow x = \sqrt{2},$$

$$b^+ = \sqrt{2} e^{-1/2} \approx 0.8578, \quad b^- = -b^+ \quad (\text{by symmetry}).$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} = \sqrt{2\pi}$$

$$h(x) = e^{-\frac{x^2}{2}}$$

$C = \{(u, v) : 0 < u < \sqrt{h(x)}\}$
 $b^- < v < b^+$

$$a = \sup_x \sqrt{h(x)} = \sup_x \sqrt{e^{-\frac{x^2}{2}}} = \sup_x e^{-\frac{x^2}{4}} = 1$$

$\downarrow$
 $x=0$

$$b = \sup_x x \sqrt{h(x)} = \sup_x \left(x e^{-\frac{x^2}{4}}\right) \Rightarrow \frac{0.8578}{\sqrt{2}} \Rightarrow b = 0.8578$$

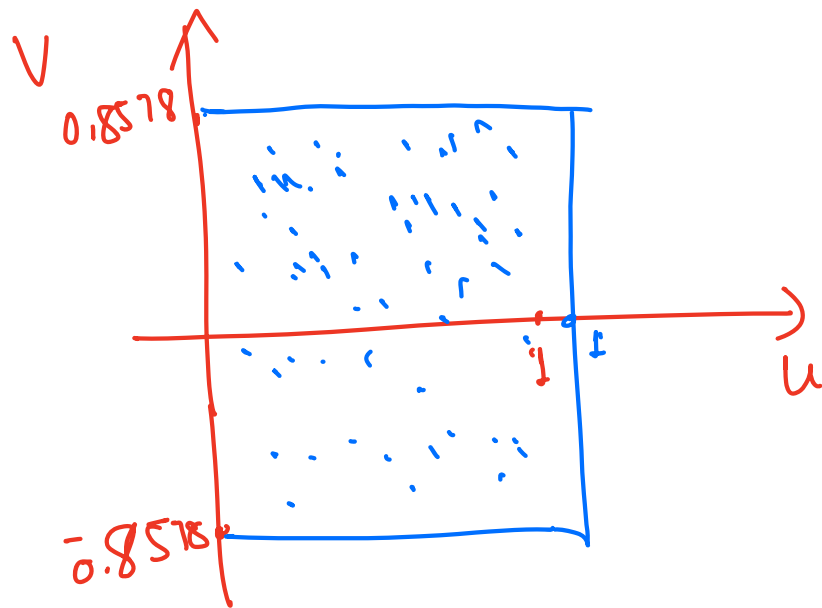
$$h(x) = e^{-\frac{x^2}{2}}$$

$$a = 1$$

$$b^- = -0.8578$$

$$b^+ = +0.8578$$

$$\Rightarrow (0, 1) \times (-0.8578, 0.8578)$$



$$\Rightarrow X = \frac{V}{u} \rightarrow U < \sqrt{h(X)}$$

otherwise

$$V \sim U_{(-0.8578, 0.8578)}$$

$$U \sim U_{[0, 1]}$$

$$\Rightarrow V_1 \sim U[-0.85, 0.85], U_1 \sim U[0, 1]$$

$$X_1 = \frac{V_1}{U_1} \rightarrow \text{check } U_1 < \sqrt{h(X)}$$

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for all x

$a =$

$U \leq 1$

$X \quad U \leq \sqrt{h(x)}$

Resulting Algorithm

1. $U \sim \text{Unif}(0, 1), V \sim \text{Unif}(-0.8578, 0.8578).$
2. $X = V/U.$
3. Accept if $U \leq \exp(-X^2/4)$, i.e. $-4 \ln U \geq X^2$; else repeat.

$$U < \sqrt{h(x)}$$
$$U < a$$

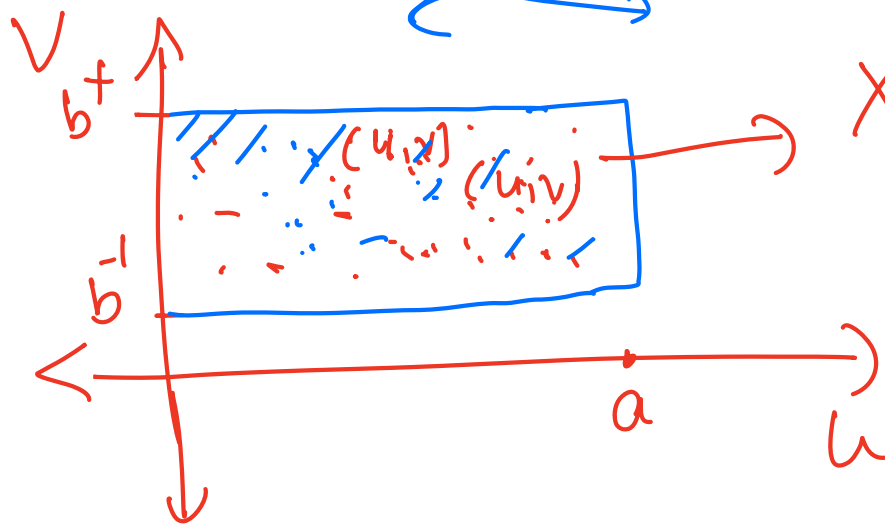
Note: No logarithm of a ratio, no trigonometry – just algebra and one exponential/log evaluation

$$\Rightarrow \frac{h(x) \geq 0}{\uparrow} \rightarrow \frac{f(x) = \frac{h(x)}{\int_{-\infty}^{\infty} h(t) dt}}{\text{pdf } f} \rightarrow$$

$$\Rightarrow \checkmark C = \left\{ (u, v) : x = \frac{v}{u}, \begin{array}{l} 0 \leq u \leq \sqrt{h(x)} \\ \bar{b} \leq v < b^+ \end{array} \right\}$$

$$U \sim U[0, \sqrt{h(x)}], \quad V \sim U[\bar{b}, b^+]$$

$$\text{Then } x = \frac{v}{u} \sim \underline{f}$$



$$X_1 = \frac{V_1}{U_1}, \quad X_2 = \frac{V_2}{U_2}$$

f

$$X_n = \frac{V_n}{U_n}$$

$\sim f$

Ratio-of-Uniforms: When to Use It

- Attractive when the density is known only up to a constant (common in Bayesian posteriors – more in Module 5).
- Often more efficient than naive acceptance–rejection because the bounding region can be tailored tightly to h .
- Generalizes naturally to some multivariate settings.
- Like acceptance–rejection, its efficiency (acceptance rate) is governed by how tightly the rectangle bounds C – the ratio $\frac{\text{Area}(C)}{\text{Area of rectangle}}$.

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Uniformity Tests

Independence and Pattern Tests

Visual Diagnostics

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Why We Must Test Our Random Numbers

- A PRNG is a deterministic algorithm – it is not *obvious* a priori that its output behaves like true i.i.d. uniforms.

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Why We Must Test Our Random Numbers

- A PRNG is a deterministic algorithm – it is not *obvious* a priori that its output behaves like true i.i.d. uniforms.
- Flawed generators have caused real, documented simulation failures (e.g. the infamous RANDU generator of the 1960s–70s, which had severe 3-D lattice structure).
- We need formal statistical tests, applied *before* trusting a generator (or a specific implementation/seed) for serious work.

Testing Uniformity

Chi-Square Goodness-of-Fit Test

Partition $(0, 1)$ into k equal bins. Let O_i = observed count in bin i out of n draws, and $E_i = n/k$ = expected count under uniformity.

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \underset{\sim}{\text{approx}} \chi_{k-1}^2 \text{ under } H_0.$$

Reject uniformity if χ^2 exceeds the critical value.

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Kolmogorov–Smirnov (KS) Test

Compare the empirical CDF $F_n(x)$ of the sample to the target CDF $F(x) = x$ on $(0, 1)$:

$$D_n = \sup_x |F_n(x) - F(x)|.$$

Large D_n (relative to its known sampling distribution) indicates a departure from uniformity.

Testing Independence / Detecting Patterns

- **Serial correlation test:** Compute the sample autocorrelation

$$\hat{\rho}_k = \frac{\sum_{n=1}^{N-k} (U_n - \bar{U})(U_{n+k} - \bar{U})}{\sum_{n=1}^N (U_n - \bar{U})^2}$$

at several lags k .

- For independent U_n , $\rho_k = 0$ for every $k > 0$.
- Hence, $\hat{\rho}_k$ should fluctuate randomly around 0.
- Significant nonzero values indicate possible dependence between successive outputs.

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 - Significant nonzero values indicate possible dependence between successive outputs.
- **Runs test:** Convert the sequence into a binary sequence, e.g.,

$$U_n < 0.5 \Rightarrow 0, \quad U_n \geq 0.5 \Rightarrow 1.$$

A *run* is a consecutive sequence of identical values.

- Under independence, the number of runs has a known expected value.
- Too few runs $\Rightarrow$ outputs tend to cluster; too many runs $\Rightarrow$ excessive alternation.

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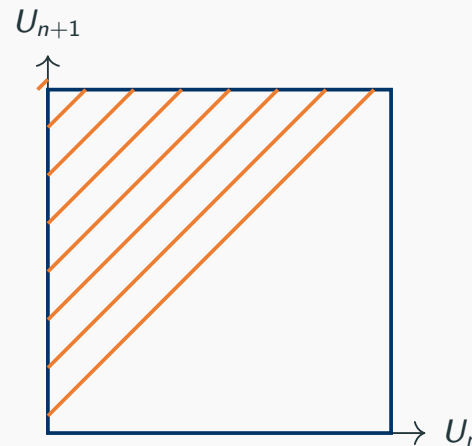
- Under independence, the number of runs has a known expected value.
 - Too few runs $\Rightarrow$ outputs tend to cluster; too many runs $\Rightarrow$ excessive alternation.
- **Spectral test:** Examine successive tuples in k dimensions

$$(U_n, U_{n+1}, \dots, U_{n+k-1})$$

- Ideally, these points should fill the k -dimensional unit cube uniformly.
- Poor generators may produce points lying approximately on a small number of parallel hyperplanes—a **lattice structure**.

Visual Diagnostics

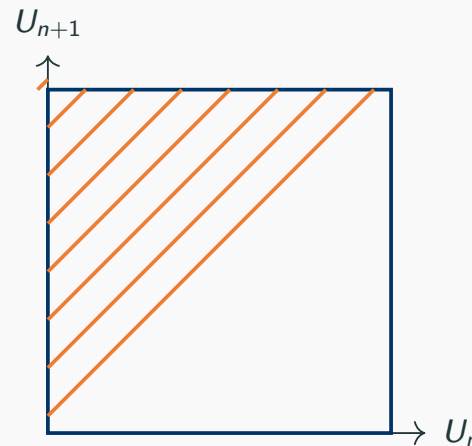
- **Histogram** of U_n : should look flat over $(0, 1)$.
- **Scatterplot of pairs** (U_n, U_{n+1}) : should fill the unit square with no visible structure.
- **QQ-plot** against the theoretical uniform quantiles: points should lie on the 45° line.



A flawed LCG: visible lattice lines

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A flawed LCG: visible lattice lines

A picture can reveal what a single test statistic misses

The infamous RANDU generator passed some 1-D tests but showed obvious planar structure in a 3-D scatterplot – a cautionary tale for always checking visually *and* formally.

Practical takeaway

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You will rarely write your own PRNG in practice – but you should know *how* to check whether the one you're relying on (R's, Python's, etc.) is trustworthy for your specific application, especially in high dimensions or for rare-event simulation.

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Putting It All Together

Method	Needs	Best suited for
Inverse Transform	Closed-form F^{-1}	Exponential, Weibull, Pareto, discrete r.v.s
Acceptance–Rejection	Envelope g , bound c	General shapes; f known up to constant
Box–Muller	—	Normal (exact, elegant)
Ratio-of-Uniforms	$h \propto f$	Unimodal densities; f known up to constant

The Big Picture

- A trustworthy **PRNG** gives us the raw stream of $U(0, 1)$ variables.
- **Transformation techniques** turn that stream into draws from any target distribution.
- We must always **diagnose randomness** (formally and visually) and be honest about **simulation error**.

Next: Module 2 – Monte Carlo Methods